

Submission Summary

Conference Name

IEEE Chandigarh Subsection International Conference

Track Name

Artificial Intelligence and Data Science

Paper ID

2132

Paper Title

Diabetes Prediction Using Supervised Machine Learning Models

Abstract

An appropriate model must be chosen when making predictions in healthcare; there should be a balance between predictive accuracy (the ability for the model to make correct predictions), computational efficiency (the speed and amount of resources required by the model), and interpretability (how well we can understand the decision process of the model). Supervised machine learning (SML) algorithms have been used extensively in the field of disease prediction, because they can learn from and identify patterns in labelled datasets from the medical field. Unfortunately, a comprehensive review to assess multiple machine learning algorithms on preprocessed and consistently evaluated datasets remains undetermined based on the current literature. The purpose of this study is to fill this knowledge gap, by conducting a thorough reviewed analysis comparing various machine learning supervised prediction models (SML) for the purpose of predicting diabetes, utilizing the PIMA Indians Diabetes Database. Four specific algorithms (Logistic Regression, Decision Tree, Support Vector Machine (SVM), and Random Forest) were evaluated with regard to performance against standardised training and testing for different data sets, including normalising of feature data, and handling of missing data. Performance was measured against 4 key criteria: accuracy, precision, recall, and F1-score. The outcome from this study indicated that the Random Forest algorithm is outperformed only slightly by SVM (Random Forest – 91% accuracy; SVM – 86% accuracy), with the Decision Tree algorithm performing at approximately 83% accuracy while the Logistic Regression algorithm performed at roughly 79% accuracy. The findings of this study indicate that using ensemble based learning models such as Random Forest provides significant improvements to prediction accuracy through eliminating overfitting and evaluating complex relationships amongst a model's features. This study will provide practical guidance for the identification of reliable and efficient SML algorithms which can be used as early detection of diabetes and assist in decision making within clinical decision making systems.

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Submission Files

Diabetes Prediction Using Supervised Machine Learning Models.pdf (1 Mb, 2026-05-14 16:40:51 GMT+5:30)

Submission Questions Response

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It is confirmed that at least one author will register in full to attend and present the paper at the Conference.

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